IMAGE PROCESSING TECHNIQUES – EXTENDED CLASS ACTIVITIES WORKSHEET

UG Level | Duration: 10 Minutes per Activity

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Activity 6: Thresholding (10 Minutes)

Given Pixel Values:

45, 60, 120, 130, 200, 210, 75, 90

Tasks:

1. Apply threshold T = 100.

The threshold is **100**.

- If pixel value **≥ 100 → 1**
- If pixel value **< 100 → 0**

2. Convert values into binary (0 or 1).

45 – 0
60 – 0

120 – 1

130 – 1

200 – 1

210 – 1

75 – 0

90 - 0

3. Write the output vector.

[0, 0, 1, 1, 1, 1, 0, 0]

4. What happens if threshold is increased?

If the threshold is increased, **fewer pixels will be classified as 1**, because more pixel values will fall below the new threshold and become 0. For example, if the threshold is increased from 100 to 150, the values 120 and 130 will also become 0. Therefore, increasing the threshold reduces the number of foreground pixels in the binary image and can cause some objects or details to disappear.